ALICE L WALTON SCHOOL OF MEDICINE

Pharmacokinetics1– 1 of 3 – Drug absorption and distribution

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Adapted from: Drug Administration, Metabolism, and Excretion

Listen to this Brick

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Predict drug bioavailability and first-pass effect, delineating advantages, and disadvantages of utilizing a particular route of drug administration.
2. Explain the factors that affect drug absorption and distribution.
3. Identify rele kinetic principles and apply those to solve clinical problems and predict outcomes.

CASE CONNECTION

MM is a 47-year-old female who underwent a vaginal hysterectomy because of severe endometriosis several years ago. She is having very bothersome hot flashes, and after trying multiple nonpharmacologic methods to control them without benefit, she requests hormone replacement therapy for relief. You agree that a few years of taking hormones is a reasonable treatment in this situation and advise MM that you'd like to prescribe an estrogen patch. "Can't I just take pills?" she asks. "That's what my mom used when she had hot flashes, and they worked well for her."

How will you explain how the pharmacokinetics of transdermal estrogen might differ from its oral counterpart? Consider your answer as you read, and we'll revisit MM at the end of the brick.

GO TO CONCLUSION

INTRODUCTION TO PHARMACOKINETICS

In the first week of pharmacology, we are going to focus on important principles of pharmacology and drug action. Two terms are going to be constantly used in this first week, Pharmacokinetics (PK) and Pharmacodynamics (PD), commonly referred as PK/PD.

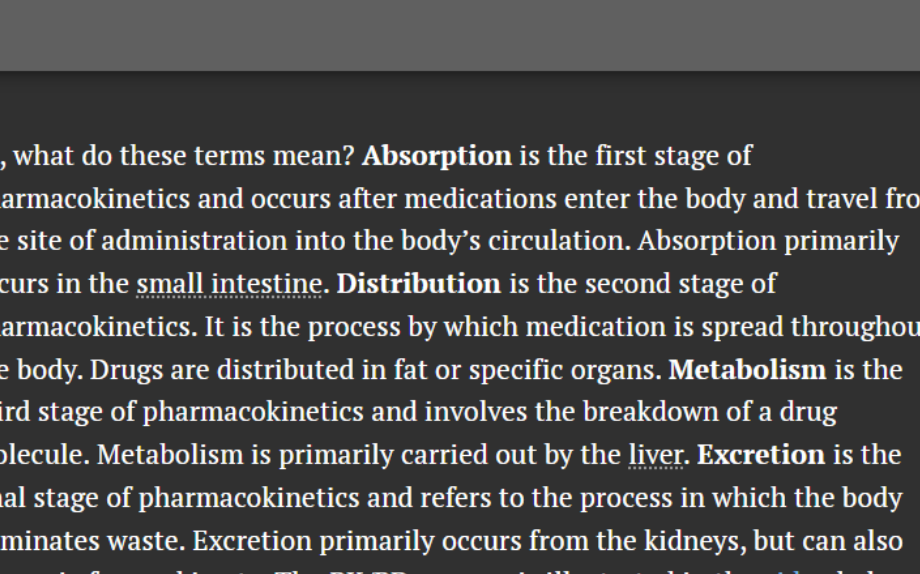
Pharmacokinetics r rug absorption, distribution, metabolism, and excretion. **Pharmacodynamics** refers to the study of biological, cellular, molecular actions through which the drug elicits its actions. In other words, pharmacokinetics is what the *human body does to a drug*, and pharmacodynamics is *what the drug does to the human body*. This might be a very simplified version of these terms, but it makes it easy to understand these concepts.

What is the difference between pharmacokinetics and pharmacodynamics?

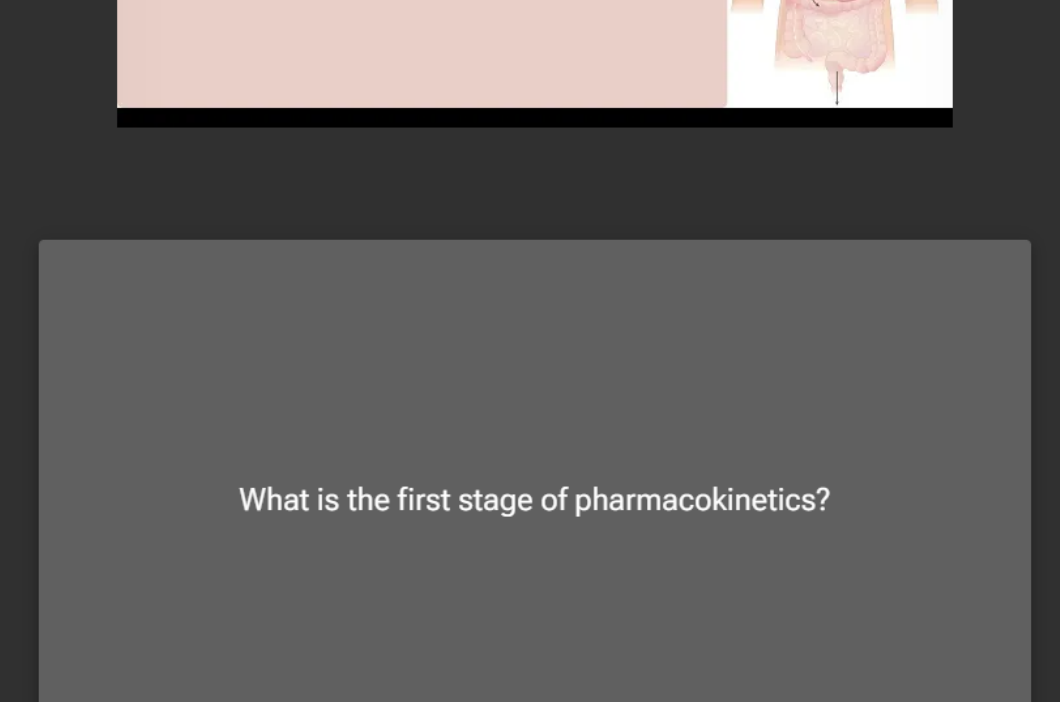
In this Brick, we will discuss some of the pharmacokinetic parameters. As pharmacokinetics of a drug is a time course, it is described in four stages, namely absorption (A), distribution (D), metabolism (M), and excretion (E) of the drug. We commonly refer to this process together as ADME, which in a four-letter word also describes the pharmacokinetics of a drug.

What does ADME stand for in pharmacokinetics?

So, what do these terms mean? **Absorption** is the first stage of pharmacokinetics and occurs after medications enter the body and travel from the site of administration into the body's circulation. Absorption primarily occurs in the small intestine. **Distribution** is the second stage of pharmacokinetics. It is the process by which medication is spread throughout the body. Drugs are distributed in fat or specific organs. **Metabolism** is the third stage of pharmacokinetics and involves the breakdown of a drug molecule. Metabolism is primarily carried out by the liver. **Excretion** is the final stage of pharmacokinetics and refers to the process in which the body eliminates waste. Excretion primarily occurs from the kidneys, but can also occur via feces, skin etc. The PK/PD process is illustrated in the video below.



What is the first stage of pharmacokinetics?


Directions: Drag the terms to the appropriate purple box then click "Submit"
Absorption
Distribution
Metabolism
Excretion
Ingestion of medicine
Submit

What are the various routes of drug administration? (LO#1)

Understanding how drugs get into the body, and once there, how much of that drug reaches systemic circulation, are the first steps in treating patients with drugs. Drugs can be administered by several different routes, which are classified as **enteral**, from the Greek word *enteron*, which means intestine, or **parenteral**, which means beside the enteron (Figure 2).

Enteral routes are oral, oral mucosa (eg, buccal), and rectal. Parenteral drug administration happens outside the gastrointestinal (GI) tract. The most common **parenteral routes** of drug administration are intravenous (IV), intramuscular (IM), intradermal, and subcutaneous injections. Other parenteral routes include transd (eardrops), nasal, vaginal, and intrathecal (spinal cord).

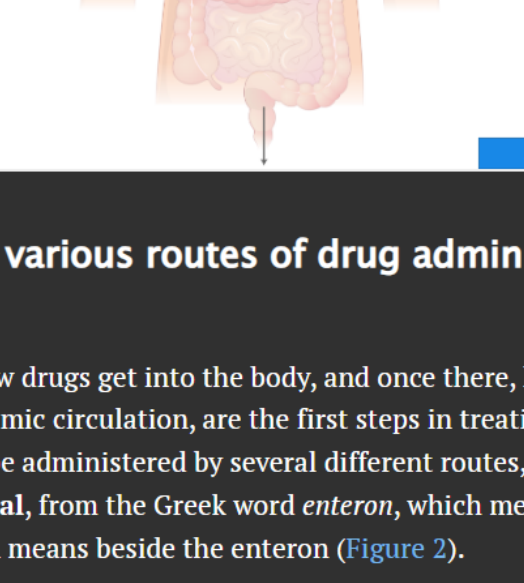


Figure 2

A medication administration route is often classified by the location at which the drug is applied, such as oral or intravenous. The choice of routes in which the medications are applied depends on convenience, the drug's properties, and pharmacokinetics. Therefore, understanding the characteristics of the various routes and associated techniques is crucial.

CLINICAL CORRELATION

Many interprofessional healthcare team members are involved in administering medications to patients. Each medication administration route has unique contraindications, and the healthcare team members must recognize them. This activity describes medication administration routes and explains the interprofessional team's role in improving care for patients who undergo medication administration.

Enteral routes of administration

Oral administration of medication is a convenient, cost-effective, and most commonly used medication administration route. The primary site of drug absorption is usually the small intestine, and the bioavailability of the medication is influenced by the amount of drug absorbed across the intestinal epithelium.

A **sublingual or buccal route** is another form of the enteral route of medication. By applying the drug directly under the tongue (sublingual) or on the cheek (buccal), the medication undergoes a passive diffusion through the venous blood in the oral cavity, which bypasses the hepatic portal vein and the liver. Compared to sublingual tissue, which has highly permeable mucosa with rapid access to the underlying capillaries, buccal tissue is less permeable and has slower drug absorption.

A **rectal route** is another enteral route of medication administration, and it allows for rapid and effective absorption of medications via the highly vascularized rectal m sorbed in the rectum directly goes to the liver.

What is the advantage of sublingual drug administration?

Parenteral routes of administration

Intravenous injection is a common parenteral route of medication administration that delivers the medications to the systemic circulation. It is indicated when a rapid drug effect is desired, a precise serum drug level is needed, or when drugs are unstable or poorly absorbed in the gastrointestinal tract. Given their superficial location on the skin, peripheral veins provide easy access to the circulatory system and are often utilized in the parenteral administration of medications. The upper extremity is usually the preferred site for intravenous medication as it has a lower incidence of thrombophlebitis and thrombosis than the lower limbs.

What is a key advantage of intravenous injection?

An **intramuscular** medication route is utilized when oral drug absorption occurs in an erratic or incomplete pattern. It can be administered in different body muscles, including the deltoid, dorsogluteal, ventrogluteal, rectus femoris, or vastus lateralis muscles. A depot preparation of the drug can be given intramuscularly, and the medication dissolves slowly into the circulation to provide a sustained dose over a more extended time.

What is the a ministration?

Subcutaneous injections are another form of the parental route of medication and are administered to the layer of skin referred to as cutis, just below the dermis and epidermis layers. This is used when the drug's molecular size is too large to be effectively absorbed in the intestinal tract or when better bioavailability or a faster absorption rate is needed than the oral route. Subcutaneous tissue has few blood vessels; therefore, the medications injected undergo absorption at a slow, sustained rate. It is easy to administer and requires minimal skills, so patients can often self-administer the medication. Common medications administered subcutaneously include insulin, heparin, and monoclonal antibodies.

Other routes of administration

Inhalation route

The alveolar epithelium is approximately 100 square meters in adults, and the alveolar volume is 4000 to 6000 mL compared to the airway volume of 400 ml. This provides a greater surface area for absorption of the inhaled substances. An inhaled medication route can be utilized to administer drugs effectively and small doses can produ rs to the absorption of substances in the a fluid including the mucus, the epithelial membrane, extravascular space, and vascular endothelium.

Transdermal route

The transdermal route of drug delivery involves applying medication directly to the skin, allowing it to be absorbed into the bloodstream. The use of patches, ointments, or creams are used for this route of delivery. The drug slowly permeates through the layers of skin in a non-invasive manner. The absorption may be slow, but the effects might last longer depending on the characteristics of the drug.

What are common forms of transdermal drug delivery?

In addition, there are other routes of administration (described in Table 1) that help localize the effect body. For instance, ocular route is used t

Table 1

ROUTE	IMPORTANT CHARACTERISTICS
Intrathecal	Injection into spinal canal or subarachnoid space to reach the central nervous system Very fast absorption
Transdermal	Achieves systemic effect through slow absorption through the skin
Topical	Local effect directly on the skin
Ocular	Drug absorption limited to eye because of physical barriers
Otic	Limited to ear disorders
Vaginal	Avoids first-pass metabolism High drug permeability May require dosing multiple times daily because of vaginal self-cleaning effect

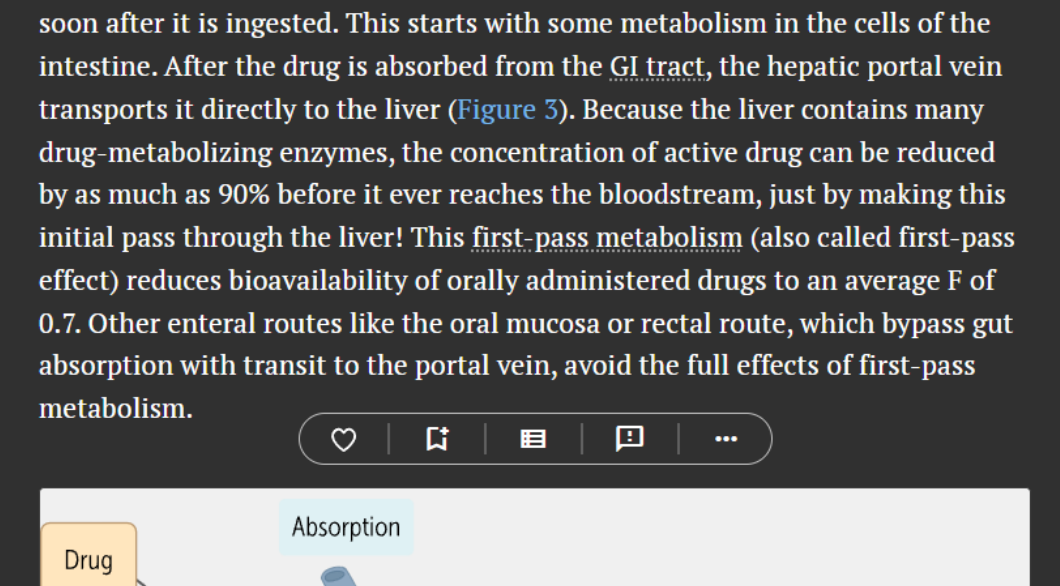
What is drug bioavailability?

Prescribing a drug doesn't mean all of it is available for action. The various routes of drug absorption can affect the amount of drug available. The **fractional bioavailability (F)** is the amount of administered drug that enters the systemic circulation. Bioavailability is reported as a percentage from 0 to 100%. F is the % of drug absorbed from the extravascular relative to the intravascular administration. Hence F is affected by the route of administration, absorption, and other factors (discussed later). As you might guess, IV injections h the drug is injected directly into the systemic circulation. This means that 100% of the absorbed drug is available. These drugs can then bind to their targets without being subjected to deactivating steps elsewhere in the body. This is why IV drug delivery yields a much more precise dose to the drug target.

Plotting the plasma concentrations of the administered drug versus time helps us calculate the **area under the curve (AUC).** Calculating the AUC helps us understand how much of the drug reaches systemic circulation and how long does it stay there. A **higher AUC** indicates more circulation over time (or greater exposure), whereas a **lower AUC** indicates poor drug absorption, rapid metabolism, or increased clearance. (we will learn about clearance in the upcoming brick).

What is First-Pass Metabolism?

Other drug administration routes have much lower F because they are metabolized or modified prior to entering the circulation. For example, a drug that is administered orally undergoes a large reduction in concentration very soon after it is ingested. This starts with some metabolism in the cells of the intestine. After the drug is absorbed from the GI tract, the hepatic portal vein transports it directly to the liver (Figure 3). Because the liver contains many drug-metabolizing enzymes, the concentration of active drug can be reduced by as much as 90% before it ever reaches the bloodstream, just by making this initial pass through the liver! This first-pass metabolism (also called first-pass effect) reduces bioavailability of orally administered drugs to an average F of 0.7. Other enteral routes like the oral mucosa or rectal route, which bypass gut absorption with transit to the portal vein, avoid the full effects of first-pass metabolism.



QUIZ

Figure 3

What factors affect drug absorption? (LO#2)

1. **pH and pka:** Majority of the drug are weak acid and/or base. Hence, the pH of the environment, concentration of protons (H+), can facilitate or impair drug absorption. Similarly, the tendency of the drug molecule to hold on to a charge, pKa can also help us predict drug absorption or excretion. For example, Penicillin G is stable within a pH range of 5-8 and can be degraded in an a).
2. **Formulation and route of administration:** Drug formulations in combination with the route of administration can affect the rate of absorption. A solid dosage oral formulation, when ingested, needs to disintegrate, and then be absorbed. Some oral formulations are immediately absorbed, while others are coated with films that can delay the absorption. For instance, a popular agent for acid reflux, omeprazole, has a coating to prevent stomach-acid-mediated degradation.
3. **Drug solubility:** The rate and extent of drug absorption depend on the drug's solubility. Poorly soluble drugs are lipophilic and have poor systemic absorption. On the other hand, freely soluble drugs are hydrophilic and often present with good systemic absorption.
4. **Age:** The age of patients also alters drug absorption. For instance, infants have incomplete development of organ and biological systems such as high gastric pH, decreased blood flow, or less intestinal absorptive area, all of which can alter drug absorption. Conversely, the elderly may have impaired systems such as less gastric emptying, decreased salivation, and altered bacterial flora, all of which can alter drug absorption.

How are drugs distributed?(LO#1 and 2)

Distribution is the process by which a drug is dispersed throughout the body's blood and tissues. Once a drug enters systemic circulation through absorption or direct administration, it moves from vascular spaces to tissues where it interacts with receptors, producing the drug's intended effect. Drugs are typically designed to target a specific receptor site, causing a predictable action. However, when a drug binds to other sites besides the target tissue, unintended actions o fects can vary from tolerable to unaccepta of the medication. For instance, while Aspirin can relieve pain, it may also cause stomach irritation as a side effect.

The distribution of a drug throughout the body is dependent on many body-related factors such as blood flow, tissue differences, plasma protein-binding, the blood-brain barrier, and the placental barrier.

Blood flow plays a crucial role in the distribution of medications throughout the body via the bloodstream. Various factors can influence blood flow and the delivery of medication, such as dehydration, atherosclerosis, uncontrolled hypertension, and heart failure. For instance, in a diabetic patient with an infected toe, blocked vessels due to atherosclerosis can hinder the antibiotic from reaching the infected area effectively.

Tissue differences also impact drug distribution. Tissues with a higher number of blood vessels, like the lungs, kidneys, liver, and brain, experience rapid distribution due to high blood flow. Conversely, tissues with fewer blood vessels, such as fat, have slower distribution. However, lipophilic drugs tend to accumulate in adipose tissue in obese individuals.

When a drug enters the bloodstream, it exists in two forms: free drug dissolved in plasma water and **protein-bound drug**. Only the free, unbound drug can pass from the bloodstream to tissues, where it interacts with receptors to produce the medication's effects. The protein-bound portion remains inactive while bound, and for many drugs, this can account for 95-98% of the total drug in the bloodstream. Protein binding can act as a reservoir, releasing the drug slowly and prolonging its action. Therefore, it's crucial to consider both the amount of free drug available to tissues and the protein-bound drug that is released over time when evaluating drug distribution.

Albumin is a key protein in the blood that affects drug binding. Factors such as malnutrition and liver disease can decrease albumin levels, leading to variations in the desired actions of medications due to changes in protein binding and distribution. Additionally, competition for plasma binding sites can impact the effects of drugs.

What is a key protein that effects drug binding?

Medications targeting the central nervous system (CNS) must overcome the significant challenge of crossing the **blood-brain barrier**, a tightly woven mesh of capillaries that protects the brain from harmful substances like poisons and viruses. Only certain lipid-based medications or those with a carrier can penetrate this barrier. Some medications, like diphenhydramine, can inadvertently cross the blood-brain barrier, causing central nervous system side effects such as drowsiness. While this side effect may be beneficial for those with sleep diffi iduals needing to stay alert during daily activities.

In *neonates and children*, the fat content is lower due to a higher total body water content. This results in decreased protein-binding capacity, and the developing blood-brain barrier allows more drugs to enter the central nervous system.

In *older adults*, even at the same body mass index, there tends to be more body fat compared to younger adults. This increased body fat can lead to a longer duration of action for many medications that accumulate in fatty tissues. Additionally, serum albumin levels decrease, resulting in more active free drug circulating within the body. Consequently, older adult clients often require lower dosages of medication due to these distribution factors.

Why do older adults often require lower medication dosages?

CLINICAL CORRELATION

While the placenta is permeable to some medications, which can potentially harm the fetus, many drugs have not been specifically studied in pregnant patients, leaving their effects on the fetus unknown. Therefore, it is essential to consider the potential impact of medications on the fetus when administering them to pregnant patients or those who may become pregnant.

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to MM, how do you explain how the pharmacokinetics of transdermal estrogen might differ from its oral counterpart?

You tell MM that oral estrogen is an option, but you prefer to use a transdermal formulation because the body reacts to it differently. When the estrogen is absorbed through the skin, it isn't changed by the liver before it enters the bloodstream, which is beneficial for reducing potential adverse effects, such as weight gain, blood clots, lipid effects, and other hormonal changes. After hearing that, MM accepts the patch as her best option.

Summary

- Pharmacokinetics (PK) involves the study of drug absorption, distribution, metabolism, and excretion (ADME).
- Pharmacodynamics (PD) refers to the biological and physiological effects of drugs on the b
- Routes of drug ad (oral, buccal, rectal) and parenteral (IV, IM, subcutaneous, etc.).
- Bioavailability is the fraction of an administered drug that reaches systemic circulation, affected by first-pass metabolism and route of administration.
- Drug distribution depends on factors like blood flow, tissue differences, plasma protein binding, and barriers like the blood-brain barrier.

Review Questions

1. Which of the following medications would most likely undergo hepatic first-pass metabolism?

A. ○ Intramuscular cefepime
B. ○ Intravenous phenobarbital
C. ● Oral estrogen
D. ○ Subcutaneous insulin
E. ○ Transdermal meperidine

Hide Explanation

The correct answer is oral estrogen (C). First-pass metabolism describes the modification of drugs either in the GI tract or after GI absorption in the first pass through the liver. It d on-oral routes, such as intramuscular cefepime (A), intravenous phenobarbital (B), subcutaneous insulin (D), or transdermal meperidine (E).

2. A 49-year-old male with gout controlled with probenecid was recently prescribed a cephalosporin antibiotic for pneumonia. Which of the following cephalosporin drug excretion pathways is inhibited by probenecid?

A. ○ Biliary excretion
B. ○ Glomerular filtration
C. ○ Passive diffusion into tubules
D. ● Renal secretion

Hide Explanation

The correct answer is renal secretion (E). Probenecid blocks the OATs, which are responsible for active secretion of drugs into renal tubules for elimination. Thus, probenecid would decrease drug excretion, resulting in increased drug levels in the blood. Biliary excretion (A) can help eliminate lipid-soluble drugs and is not affected by probenecid. Glomerular filtration (C) is a form of renal drug excretion of unbound drugs but is not affected by probenecid. Passive diffusion into tubules (D) is another way that drugs are renally excreted and is not affected by probenecid.

3. Which form of a drug can pass from the bloodstream to tissues?

A. ● Free drug
B. ○ Protein bound drug
C. ○ Dissolved drug
D. ○ Metabolized drug
E. ○ Bound drug

Hide Explanation

The correct answer is free drug (A). Only the free, unbound drug can pass from the bloodstream to tissues and interact with receptors. Protein-bound drug (B) remains inactive while bound. Dissolved drug (C) is not a specific term used in this context. Metabolized drug (D) refers to the drug after it has been processed by the body. Bound drug (E) is another term for protein-bound drug, which remains inactive.

4. Why do older adults often require lower medication dosages?

A. ○ Higher total body water
B. ○ Decreased protein-binding capacity
C. ● Increase
D. ○ Increased serum albumin

Hide Explanation

The correct answer is increased body fat (C). Older adults have more body fat, leading to longer drug action. Higher total body water (A) is more relevant to neonates and children. Decreased protein-binding capacity (B) applies to neonates and children. Increased serum albumin (D) is incorrect as serum albumin levels decrease in older adults.

5. Which route of administration has a bioavailability (F) of 100%?

A. ○ Oral
B. ● IV injection
C. ○ Subcutaneous injection
D. ○ Intramuscular injection
E. ○ Inhalation

Hide Explanation

The correct answer is 1.0 (100%) because the drug is injected dire administration (A) and the drug is injected directly into the systemic circulation. Oral administration (A) and subcutaneous injection (C) have lower bioavailability due to absorption barriers. Intramuscular injection (D) and inhalation (E) also have bioavailability less than 100% due to absorption and metabolic factors.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Predict drug bioavailability and first-pass effect, delineating advantages, and disadvantages of utilizing a particular route of drug administration.~~
02 ☑ ~~Explain the factors that affect drug absorption and distribution.~~
03 ☑ ~~Identify relevant pharmacodynamic and pharmacokinetic principles and apply those to solve clinical problems and predict outcomes.~~

Objective Confidence: 3/3